

CS112 Data Structures and Algorithms COURSE OUTLINE

1. **SEMESTER/YEAR:** *Semester 2, 2026*
2. **MODE OF DELIVERY/LOCATION:** *F2F at Laucala, and Blended for all campuses*
3. **PRE-REQUISITES:** *CS111*

4. **COURSE CO-ORDINATOR**

Name: Dr. Anurag Sharma
Office: Level 3, ICT Building A, Japan Pacific ICT Center
Phone: 3232618
Email: anuraganand.sharma@usp.ac.fj
Labs: Wed 11-1 (online) for blended students
Consultation Hours: Monday 12-1

5. **TEACHING TEAM**

Name: Kaushik Kumar
Email: kaushik.kumar@usp.ac.fj
Labs: Mon 9-11 (NL), Mon 3-5 (TL3), Wed 9-11 (TL4), Thurs 1-3 (NL), Thurs 3-5 (TL4), Fri 5-7 (NL)

Name: Krish Rao
Email: s11219601@student.usp.ac.fj
Labs: Mon 5-7 (NL), Fri 3-5 (NL)

Name: Rudr Prasad (at-risk tutor)
Email: s11219309@student.usp.ac.fj
Labs: Fri 3-4 (TL1B and online) from Week 3

6. **LECTURE TIMES & VENUE**

Day	Time	Location
Monday	2pm	018-001
Wednesday	2pm	010-111
Friday	1pm	092-003

7. EMERGENCY CONTACT

Name: Dr. MGM Khan, Deputy Head of School (Learning, Teaching and Quality)
Email: mgm.khan@usp.ac.fj
Phone: (+679) 32 32507

8. COURSE DESCRIPTION

This course provides the necessary skills in the development of software applications via C++ programming language. However, the knowledge acquired can be applied to develop a software application independent of the programming language. In this course students will learn the syntax, logic and operation of fundamental data structures like arrays, queues, stacks and trees. Some common operations include basic algorithms like searching, sorting and recursion. Students will be able to appreciate the importance of developing software in computationally efficient way by choosing appropriate data structures and algorithms. This course focuses on the development of software using the object-oriented paradigm.

9. COURSE LEARNING OUTCOMES

On successful completion of this course, students should be able to:

1. Demonstrate the usage of various data structures in programming.
2. Develop data structures using object-oriented programming to solve well-defined computing problems.
3. Develop algorithms for small-sized problems using object oriented design.

10. PROGRAM GRADUATE OUTCOMES

- Bachelor of Software Engineering (BSE) Graduate Outcomes
"The graduates should be able to develop real-world software systems".
- Bachelor of Arts (BA) & Bachelor of Science (BSc)
"Provide solutions to real-world Information Communication Technology (ICT) problems."
- Bachelor of Networks and Security (BNS)
"The graduates should be able to design and implement secured networks to communicate locally as well as globally"

11. USP GRADUATE OUTCOMES

The USP graduate outcomes are as follows:

1. **Communication:** Graduates will be able to communicate ideas clearly and persuasively in structured formats using language and other modes of communication that are appropriate for context, audience and specific disciplinary conventions.

2. **Creativity:** Graduates will extend boundaries of current knowledge in a particular discipline or profession.
3. **Critical thinking and quantitative reasoning:** Graduates will be able to evaluate multiple perspectives and arrive at a reasonable independent judgement based on evidence.
4. **Ethics:** Graduates will demonstrate a commitment to high ethical standards in scholarly, professional and socio-cultural practices.
5. **Pacific consciousness:** Graduates will recognize the cultural heritage and diversity of Pacific societies for sustainable development in a contemporary environment.
6. **Professionalism:** Graduates will demonstrate the ability to carry out tasks to acceptable standards within their profession and occupations.
7. **Teamwork:** Graduates will work together in a respectful and collaborative manner to complete tasks within teams to achieve an outcome.

12. COURSE CONTENT

Focus for Each Week	
1	Revision of CS111 materials: Arrays, Functions, and Strings
2	Arrays
3	C++ Pointers & Introduction to Structs
4	Data Abstraction and Introduction to Classes and objects
5	Object-Oriented Concepts: inheritance and polymorphism
6	Processors & linking and memory allocations
7	Recursion
8	Algorithms and Introduction to complexities
9	Dynamic memory allocation
10	Linked Lists
11	Stacks & Queues
12	Binary Trees
13	Standard Template Library (STL) and Handling exceptions. Hash tables (optional)
14	Study Break
15	Exam
16	Exam

13. PRESCRIBED TEXT AND OTHER RESOURCES

Prescribed Text [optional]

Malik, D. S., (2017). *C++ Programming: Program Design Including Data Structures*, 8th Edition, Cengage Learning US, ISBN: 9781337117562

References

Horstmann, C., (2010). *C++ for Everyone*, 2nd Edition, Wiley, ISBN: 9780470927137

Open Educational Resource (OER)

Morin, P., (2015). Open Data Structures: An Introduction, Athabasca University, <https://www.oercommons.org/>

Students are also advised to use search engines like google to learn more about programming or get examples. Here is a recommended YouTube Channel:

https://www.youtube.com/watch?v=2c_Pfv4Ujmc&t=86s

14. ALIGNMENT OF LEARNING OUTCOMES, ACTIVITIES AND ASSESSMENT

The following table demonstrates the alignment of the course learning outcomes to appropriate activities and assessments and the links to programme and USP graduate outcomes.

Course Learning Outcome	Associated Teaching and Learning Activities	Assessment	Programme Outcomes	USP Graduate Outcomes
1	Concepts are explained in lectures and practiced in labs	Assignment 1 Mid Sem Test Final Exam	Programming Teamwork	Teamwork
2	Concepts are explained in lectures and practiced in labs	Assignment 2 Final Exam	Data and Information Management Abstraction	Critical Thinking
3	Concepts are explained in lectures and practiced in labs	Assignment 2 Final Exam	Programming concepts/issues	Critical Thinking

15. ASSESSMENT**15.1 ASSESSMENT PORTFOLIO**

TYPE OF ASSESSMENT	WEIGHT	COMMENTS	LEARNING OUTCOME	AI USAGE POLICY
14.1.1 CONTINUOUS ASSESSMENT – 60%				
Programming Assignments	35	2 programming assignments are required to be done by student during the semester.	All	1
Mid-Semester Test	20	One short test performed under strict supervision with an allocated time of 50 minutes. This will test students' ability to analyze parts of a real-world	1,2	1

		problem and suggest an implementation by writing algorithm or code as required.		
Tutorial Participation	5	This activity will assess lecture and lab concepts taught during the lab. F2F and blended students must participate in class discussion during the lab/tutorial.	1,2	2
14.1.2 FINAL EXAMINATION – 40%				
Final Exam	40	This examination will consist of a three (3) hour paper at the end of your course. You must score a minimum of 40% in this exam in order to pass the course.	All	1

15.2 PASSING THE UNIT

In order to be awarded a pass in this unit, students must:

- Achieve an overall mark of at least 50% in the course.
- Achieve at least 40% in the final exam.
- Attend at least 60% of Labs (F2F students only).
- If you DO NOT comply with these criteria you are automatically deemed to have failed the course although you will be allowed to attempt all assessments.

15.3 GRADE DISTRIBUTION

A+	A	B+	B	C+	C	D	E
85-100	78-84	71-77	64-70	57-63	50-56	40-49	0-39

15.4 ASSESSMENT POLICIES AND REGULATIONS

For detailed regulations, please refer to the USP Handbook and Calendar.

Plagiarism, copying materials from other sources without proper referencing and acknowledge of the source is a serious offence and will be dealt with severely. In the **Regulations Governing Academic Misconduct** section of the USP Handbook & Calendar plagiarism is defined as “the copying of another person’s creative work and using it as one’s own – without explicitly giving credit to the original creator. Work copied without acknowledgement from a book, from another student’s work, from the internet or from any other source”. If lecturer is satisfied that plagiarism has occurred, they will report the matter to the Head of School. They can reduce marks appropriately. If the matter is seen as serious enough it can be taken to the Student Disciplinary Committee by the Head of School.

AI Usage Policy

1. **No AI Usage:** AI tools cannot be used in the assessment task in this course. In this assessment, you must NOT use generative AI to generate any materials or content in relation to the assessment task.
2. **Selective AI Usage:** AI tools are restricted for certain functions in this assessment task. The detailed instructions for using selective AI usage will be provided in the assessment. Any use of generative AI must be appropriately acknowledged.
3. **Open AI Usage:** AI tools are not restricted for this assessment task. In this assessment, you can use generative AI to assist you. Any use of generative AI must be appropriately acknowledged.

16. IMPORTANT DATES

Activity	Week	Weighting (%)
Assignment 1	5	15
Mid-Semester Test	8	20
Assignment 2	9	20
Tutorial Activity	2-13	5

17. ADDITIONAL COURSE ATTENDANCE

17.1 LABORATORY / TUTORIALS

Attendance in labs is mandatory. Students can choose their lab sessions by signing up on the course shell on Moodle. All labs start from the second week. No tutorials. Tutorials to be part of Labs.

17.2 ONLINE WORK

NA

18. STUDENT SUPPORT

18.1 ONLINE HELP & e-Learning INFORMATION

All course materials that will be made available on Moodle, students are encouraged to print out notes well before time and read through it at least once so that it makes sense once lectures are delivered.

All information and material will be provided on Moodle share for CS230 Course in the website: <http://elearn.usp.ac.fj>. You may login using your SOLS username and password.

Students may either call in personally or seek help via email. Any student seeking help relating to the course via email should include the following:

18.2 CFL STUDENT LEARNING SUPPORT (SLS) SERVICES

The University provides additional academic assistance through the **Student Learning Support (SLS)** unit within the Centre for Flexible Learning (CFL). SLS equips students with essential academic and study skills to help them meet course requirements.

A range of free services is available to support students in managing the demands of tertiary education at USP. These services include:

- **Drop-In Consultations – Face to Face & Online:** Students may meet with an SLS Specialist at SLS Hubs or staff offices to discuss their academic needs. Support includes assignment unpacking and guidance on study skills such as time management, active learning strategies, and exam preparation.
For assignment unpacking, students are advised to book their initial consultation at least two weeks before the due date. Appointments can be arranged by emailing the relevant SLS staff member to schedule a mutually convenient time. Students can access the same support services online via the generic email: learning.support@usp.ac.fj. Responses are typically provided within two business days or sooner.
- **Academic Skills Workshops – Generic & Customized:** Students are strongly encouraged to engage with a variety of self-paced workshops available on the SLS Moodle page. These workshops are accessible 24/7 and include comprehensive notes and activities for each topic. Topics include time management, critical reading and writing skills, avoiding plagiarism, and more. Workshops may also be offered at the request of course coordinators and customised to meet the needs of specific courses and students.
- **Peer Assisted Study Sessions (PASS) – Face to Face & Online:** These sessions are facilitated weekly by high-achieving senior students and focus on interactive group activities that strengthen problem-solving skills, study techniques, and assessment preparation. The service is offered for selected courses only.
- **Dedicated Literacy and Numeracy Support** – Students can access specialised support in literacy and numeracy through both face-to-face and online consultations with dedicated staff.
- **Student Learning Support (SLS) – MOODLE Page** – All registered students have free access to a dedicated MOODLE shell which provides:
 - SLS staff contacts and SLS Hub locations at Fiji and Regional campuses
 - Peer Assisted Study Sessions (PASS) timetables and resources for Schools
 - Specialized learning support for English Language Skills (ELS) and Numeracy
 - Just-in-time self-paced workshops on a wide range of academic, study, ICT, and other generic skills

- **Dedicated At-Risk and Disability Support** – The SLS team collaborates with relevant sections of the University to assess student needs and provide tailored academic support, as required.

Reach out to a Learning Specialist Now:

SLS Specialist	Email	Phone	Location
Prithvi Narayan	prithvi.narayan@usp.ac.fj	32 32264	FSTE Building (Top Floor, Admin Space) Laucala Campus
Vrinda Singh	vrinda.singh@usp.ac.fj	32 30358	
Imanuel Thakur	imanuel.thakur@usp.ac.fj	32 37167	FSTE Building (Basement Level) – SLS Hub
Ram Singh Jnr	ram.singhjnr@usp.ac.fj	32 37167	Laucala Campus

For regional campuses, please reach out to any specialist, and we will direct you to them accordingly.

19. EXPECTATIONS OF STUDENTS

- Contribute equally to the team
- Be present in lecture and lab
- Complete all the quizzes
- Spend appropriate hours of work for this course as advised in student workload.

19.1 STUDENT RESPONSIBILITIES

Students are expected to:

- Spend enough time on Moodle and other activities required in the course
- *Attend all lectures, workshops, team meetings etc. unless hampered by illness in which case you must inform your course coordinator and team members.*
- Arrange with the team members to work in their own times on the project.
- Maintain proper attire for the laboratory, clothing, shoes etc.
- *Facilitate the course coordinator with any proof that he/she was sick and unable to attend classes or assessment at the earliest possible time to avoid getting a zero mark for the missed assessment.*
- Report any damages to the equipment immediately to the user assistant or tutor.
- Be responsible for any damages of USP property due to negligence. (Negligence will be assessed by the Chief Technician(s) and the Division Coordinator(s).

- Follow all guidelines stipulated in the School's OHS regulations. (Guidelines will be provided during the first lab session.)

19.2 STUDENT WORKLOAD

Type	Hours	Comments
Lab/Tutorial Participation	24	Thirteen 1 hour 50-minute tutorials/labs
Lecture/Lab Preparation/Weekly Readings	39	Expectations are from students to devote these times reading course related materials such as lecture notes, textbooks and prepare for labs.
Lectures	39	Three lectures per week plus the time spent on doing the test.
Assignments	40	twenty hours to be spent on each assignment.
Test Participation	1	One-hour mid-semester test.
Test Preparation	15	Hours to be spent preparing for the test.
Final Exam Participation	3	A 3-hour online examination at the end of the semester.
Final Exam Preparation	30	The student is expected to spend about 20 hours preparing for the final examination
Total	191 Hours	